RT RootTrace 8D

From customer complaint to evidence-backed 8D report in minutes

Deterministic SPC + AI on AMD + engineer sign-off

AI inference on AMD GPUs · Token Factory

MIT · sample data fictional

THE PROBLEM

A rejected part starts a race against the clock

- The customer demands **containment within 24 h** and an **8D report** within days
- The evidence is scattered: the complaint email, photos, SPC sheets, the maintenance log
- Finding **when** and **why** the process changed is manual detective work
- Containment is often guessed from the lots the customer named, so more defective parts escape

Cost of getting it wrong

Line stops at the customer, sorting at both plants, chargebacks, a lower supplier rating. Each 8D also takes hours to days of engineer time.

THE USER

Supplier quality engineers

Who

Quality engineers at Tier-1 and Tier-2 automotive and industrial component suppliers (IATF 16949 plants)

Job to be done

Contain fast, prove the root cause with data, and write an 8D the customer accepts

Today

Excel, paper SPC sheets, email threads and a blank 8D template

"I don't need a chatbot. I need to know **which lots to stop** and **why**, and I need numbers I can defend in front of the customer."

THE SOLUTION

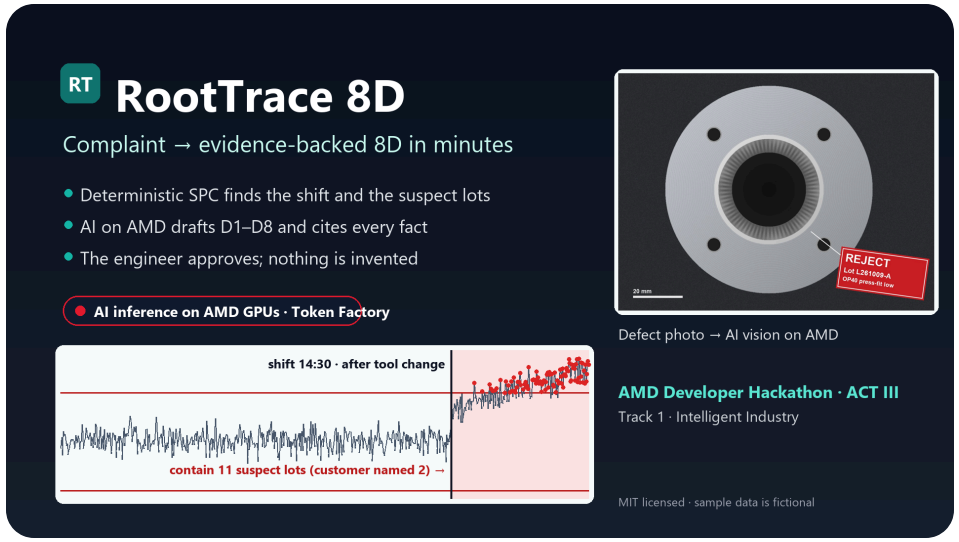
One click, eight steps, every claim cited

#	STEP	ENGINE
1–3	Read inputs → **SPC** (I-MR limits, Western Electric rules, Cp/Cpk, change point, containment) → correlate with the maintenance log	deterministic
4	Extract complaint facts with verbatim quotes	AI on AMD
5	Classify the defect photo (vision)	AI on AMD
6	Rank 6M root-cause hypotheses with supporting and contradicting evidence	AI on AMD
7	Draft the 8D, D1–D8. D3 containment is fixed to the computed suspect lots	AI on AMD
8	Number-trace check: flags any figure not found in the evidence	deterministic

Then the engineer edits and approves each section and exports a PDF. Unapproved sections print as DRAFT.

LIVE DEMO · SAMPLE CASE

The customer named 2 lots. The data says 11.



10′
shift detected 10 min after the boring-bar change

11
suspect lots to contain (customer named 2)

2.12 → −0.14
Cpk before → after the shift

100%
numbers in the draft traced to evidence

HOW IT RUNS ON AMD

One client, AMD endpoints, full trace

- An OpenAI-compatible client pointed at the **AMD TokenFactory** API, or at **vLLM on AMD Instinct MI300X** (ROCm, AMD Developer Cloud)
- A vision step on a separate AMD endpoint, with **Gemma** preferred
- Respects gateway limits: 8 concurrent requests, a 20 requests/min budget, Retry-After and backoff
- A recorded AMD run keeps the demo green under load, clearly labeled

AMD inference trace, shown in the UI

For every AI step: provider · model · live/cached · latency · tokens in/out · attempts. Judges and users can see exactly what ran on AMD.

TRUST BY DESIGN

An AI draft the engineer can defend

Evidence IDs

Every fact gets an ID (S4, L6, P2…). Hypotheses and sections must cite them, and invented IDs are dropped.

Numbers are computed

SPC numbers come from code. A trace check flags any number the AI wrote that is not in the evidence.

Human sign-off

Each section goes from AI draft to edited to approved. An edit after approval needs re-approval, and DRAFT is watermarked on export.

Uncertainty is shown: confidence levels, contradicting evidence (for example, the coolant dip came **32 h after** the shift), and "not stated" facts.

MARKET & BUSINESS MODEL

Every certified supplier plant writes 8Ds

- **Beachhead:** automotive Tier-1 and Tier-2 plants (IATF 16949), then industrial and aerospace suppliers
- **Pricing:** SaaS per plant per month. An on-premises edition runs on the plant's own AMD Instinct or Ryzen AI hardware for data-sensitive customers
- **Expansion:** quality engineer → plant quality manager → group-wide supplier quality

Sizing (assumptions to validate)

SAM ≈ plants × price × 12. For example, 10,000 EU + NA supplier plants × €400/month ≈ **€48M ARR**. Value per plant: fewer escapes and engineer hours saved on every 8D.

COMPETITION

Why RootTrace wins

	QMS SUITES	SPC SOFTWARE	GENERIC AI CHAT	ROOTTRACE 8D
Finds the shift and the suspect lots	✗	charts only	✗	✓ computed
Drafts the 8D narrative	template	✗	✓ but invents numbers	✓ evidence-cited
Reads photos and complaint emails	✗	✗	partly	✓
Human approval and audit trail	✓	✗	✗	✓
Runs on your own AMD hardware	–	–	✗	✓

ROADMAP & TEAM

Next 90 days

- MES and QMS connectors for live SPC feeds
- Multi-characteristic and subgrouped charts
- A library of past 8Ds to detect recurring causes
- Export to customer supplier-portal formats
- Pilot with 2–3 supplier plants

Team

[Your name] · builder
Built with Spec-Driven Development: constitution → spec → plan → tasks → implementation